

filters and feedback

1 Filters

- convolutions and transfer functions
- amplitude gain and phase shift

2 An Application

- transfer functions of circuits
- making a Bode plot

3 Feedback

- credit card balance
- design in the frequency domain
- paying off a credit card balance

MCS 472 Lecture 34
Industrial Math & Computation
Jan Verschelde, 8 April 2026

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filters as plants

Definition (plant)

A *plant* is a linear, time invariant, causal map, which maps signals into signals.

Examples: filters, mechanisms, circuits.

By the linearity, time invariance, and causality, the output $y = h \star u$ where h is the response to the unit impulse and u is the input.

Definition (convolutions of functions)

Let f and g be functions of t . The *convolution* $f \star g$ of f and g is

$$f \star g = \int_0^t f(\tau)g(t - \tau)d\tau.$$

linear, causal, time invariant filters

Exercise 1:

Is the filter $y(t) = u(t/2)$ linear, causal, and time invariant?

Justify your answer.

the Laplace transform

Definition (the Laplace transform)

Given an integrable function $f(t)$ in time t ,

the *Laplace transform* of $f(t)$ is $\mathcal{L}\{f\} = \int_0^{\infty} f(t)e^{-st}dt = F(s)$.

Applied to filters:

- $u(t)$ is the input signal, $y(t)$ is the output signal,
- $h(t)$ is the response of the unit impulse.

The convolution $y = h \star u$ is transformed into

$$Y(s) = H(s)U(s), \quad U = \mathcal{L}\{u\}, \quad H = \mathcal{L}\{h\}, \quad Y = \mathcal{L}\{y\},$$

where $H(s)$ is *the transfer function* of the filter.

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amplitude gain and phase shift

the continuous analogue of the theorem of Lecture 7

Theorem (gain and phase shift of a filter)

Let $H(s)$ be the transfer function of a filter.

The response to $u(t) = \sin(\omega t)$ is $y(t) = r \sin(\omega t + \phi)$, where

- $r(\omega) = |H(\omega i)|$ is the amplitude gain, and
- $\phi(\omega) = \arg H(\omega i)$ is the phase shift.

In applying the theorem:

- For $\omega = 2\pi n$, the input signal $u(t) = \sin(\omega t)$ runs at n Hertz.
- Complex numbers appear, the imaginary unit is $i = \sqrt{-1}$.
- $H(\omega i) \in \mathbb{C}$ and $H(\omega i) = re^{i\phi}$ is its polar representation.

working with complex numbers, $i = \sqrt{-1}$

Adding up

$$\begin{aligned} e^{i\omega t} &= \cos(\omega t) + i \sin(\omega t) \\ - \left(e^{-i\omega t} &= \cos(\omega t) - i \sin(\omega t) \right) \end{aligned}$$

leads to

$$e^{i\omega t} - e^{-i\omega t} = 2i \sin(\omega t).$$

Why this is useful:

$$\sin(\omega t) = \frac{1}{2i} \left(e^{i\omega t} - e^{-i\omega t} \right) \Rightarrow \text{we look at the response of } e^{i\omega t}.$$

the response of $e^{i\omega t}$

$$\begin{aligned}y(t) &= h(t) \star u(t) \\&= \int_0^t h(\tau) e^{i\omega(t-\tau)} d\tau \\&= e^{i\omega t} \int_0^t h(\tau) e^{-i\omega\tau} d\tau \\&= e^{i\omega t} \left[\int_0^\infty h(\tau) e^{-i\omega\tau} d\tau - \int_t^\infty h(\tau) e^{-i\omega\tau} d\tau \right] \\&\quad \text{Observe: } \int_0^\infty h(\tau) e^{-s\tau} d\tau = \mathcal{L}\{h\} = H(s). \\y(t) &= e^{i\omega t} H(i\omega) + o(1)\end{aligned}$$

The $o(1)$ denotes a function with limit 0 as $t \rightarrow \infty$.

the response of $\sin(\omega t)$

Apply $y(t) = e^{i\omega t} H(i\omega)$, for $u(t) = e^{i\omega t}$, to $\sin(\omega t) = \frac{1}{2i} (e^{i\omega t} - e^{-i\omega t})$:

$$\begin{aligned} h(t) \star \sin(\omega t) &= h(t) \star \frac{1}{2i} (e^{i\omega t} - e^{-i\omega t}) \\ &= \frac{1}{2i} [e^{i\omega t} H(i\omega) - e^{-i\omega t} H(-i\omega)] \end{aligned}$$

$$H(i\omega) \in \mathbb{C} : H(i\omega) = re^{i\phi} :$$

$$\begin{aligned} &= \frac{1}{2i} [re^{i(\omega t + \phi)} - re^{-i(\omega t + \phi)}] \\ &= r \sin(\omega t + \phi). \end{aligned}$$

Q.E.D.

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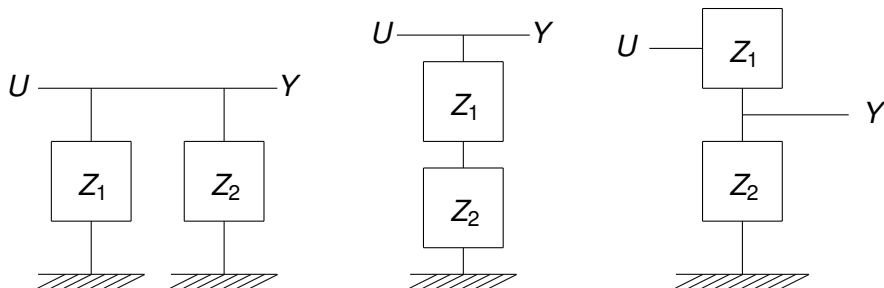
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transfer functions of circuits

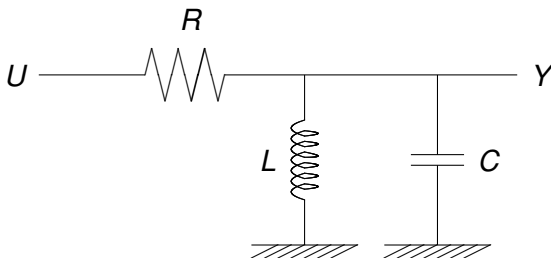
The circuits



have transfer functions, respectively:

$$\frac{Y}{U} = \frac{1}{1/Z_1 + 1/Z_2}, \quad \frac{Y}{U} = Z_1 + Z_2, \quad \text{and} \quad \frac{Y}{U} = \frac{Z_2}{Z_1 + Z_2}$$

a parallel resonant circuit



Resistor: $V/I = R$, Inductor $V/I = Ls$, Capacitor: $V/I = \frac{1}{Cs}$.

The transfer function is

$$\frac{Y}{U} = \frac{Z_2}{Z_1 + Z_2}, \quad Z_1 = R, \quad Z_2 = \frac{1}{\frac{1}{Ls} + Cs}.$$

the ordinary differential equation

Expanding the transfer function

$$\frac{Y}{U} = \frac{Z_2}{Z_1 + Z_2}, \quad Z_1 = R, \quad Z_2 = \frac{1}{\frac{1}{L}s + Cs}$$

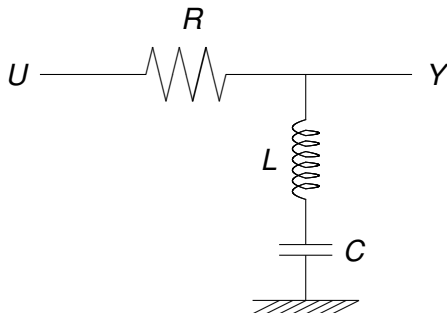
into

$$\frac{\frac{Ls}{LCs^2+1}}{R + \frac{Ls}{LCs^2+1}} = \frac{Ls}{RLCs^2 + R + Ls} = \frac{\frac{1}{RC}s}{s^2 + \frac{1}{RC}s + \frac{1}{LC}}$$

corresponds to the ordinary differential equation

$$y''(t) + \frac{1}{RC}y'(t) + \frac{1}{LC} = \frac{1}{RC}u'(t).$$

another realization of a filter



Exercise 2: Consider the above circuit.

- 1 Derive the transfer function for this circuit.
- 2 Write the corresponding ordinary differential equation.

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amplitude gain and phase shift of a filter

The transfer function is

$$H(s) = \frac{\frac{1}{RC}s}{s^2 + \frac{1}{RC}s + \frac{1}{LC}}.$$

Application of the theorem:

- $r(\omega) = |H(\omega i)|$ is the amplitude gain, and
- $\phi(\omega) = \arg H(\omega i)$ is the phase shift.

Two parameters:

- 1 $RC = 100/3$
- 2 $LC = 1/9$

code to make the Bode plot

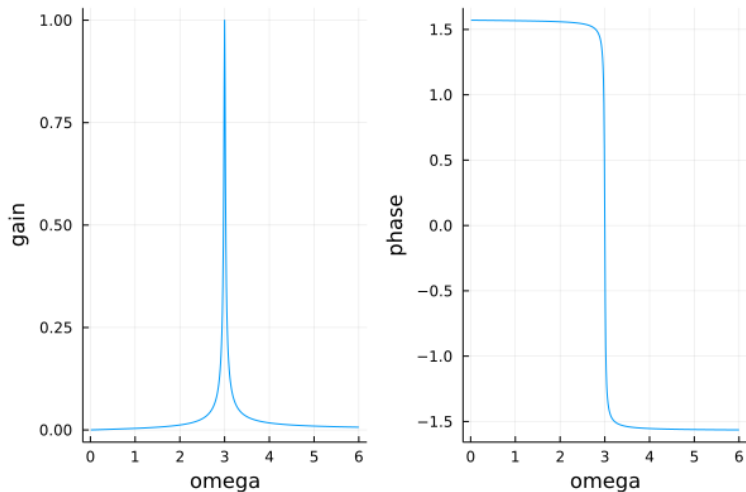
```
RC = 100/3; LC = 1/9
H(s) = ((1/RC)*s)/(s^2 + s/RC + 1/LC)
I = Complex(0, 1)

maxomega = 6
omegas = 0.01:0.01:maxomega

gains = [abs(H(I*w)) for w in omegas];
phases = [angle(H(I*w)) for w in omegas];

gainplot = plot(omegas, gains, xlabel="omega",
                ylabel = "gain", label = "")
phaseplot = plot(omegas, phases, xlabel="omega",
                 ylabel = "phase", label = "")
plot(gainplot, phaseplot)
```

the Bode plot



We call this filter a *peak filter*.

make a Bode plot

Exercise 3: Consider the transfer function

$$H(s) = \frac{s^2 + 1}{s^2 + s + 1}.$$

Make the Bode plot for ω ranging between 0 and 0.5.

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credit card balance

Consider the balance of a credit card:

- B_0 is the initial balance,
- r is the annual interest rate,
- time t is measured in years.

In the time domain: $B(t) = B_0 e^{rt}$ as the solution of

$$\frac{d}{dt}B(t) = rB(t), \quad B(0) = B_0.$$

Rewriting the ordinary differential equation:

$$\frac{d}{dt}B(t) - rB(t) = 0 \quad \Leftrightarrow \quad (D - r)B = 0, \quad D = \frac{d}{dt}.$$

In the frequency domain: $P(s) = \frac{B_0}{s - r}$ is the transfer function.

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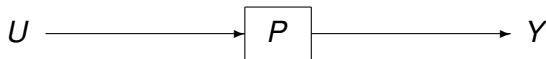
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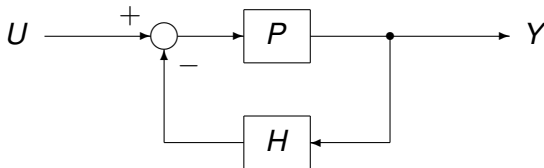
an open loop system

For $r > 0$, $P(s) = \frac{B_0}{s-r}$ is unstable.



Feedback can be used to stabilize a plant.

a closed loop system



The transfer function $G(s)$ of the closed loop system is

$$G(s) = \frac{P(s)}{1 + P(s)H(s)},$$

as the input to P is

$$U(s) - H(s)Y(s)$$

and then

$$Y(s) = P(s)(U(s) - H(s)Y(s)) \Rightarrow (1 + P(s)H(s))Y(s) = P(s)U(s).$$

an example of proportional feedback

Consider $P(s) = \frac{1}{s-1}$.

A linear amplifier with gain κ has transfer function $H(s) = \kappa$.

The transfer function of the closed loop system is

$$G(s) = \frac{P(s)}{1 + P(s)H(s)} = \frac{1/(s-1)}{1 + \kappa/(s-1)} = \frac{1}{s + \kappa - 1}.$$

The closed loop system is stable for any $\kappa > 1$.

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paying off a credit card balance

Stabilize $P(s) = \frac{B_0}{s-r}$ with proportional feedback κ .

$$\begin{aligned} G(s) &= \frac{P(s)}{1 + \kappa P(s)} \\ &= \frac{\frac{B_0}{s-r}}{1 + \kappa \frac{B_0}{s-r}} \\ &= \frac{B_0}{s-r + \kappa B_0} \\ &= \frac{B_0}{s - (r - \kappa B_0)} \end{aligned}$$

The root of $G(s)$ is $r - \kappa B_0$. For stability: $r < \kappa B_0$, or $\kappa > r/B_0$.

a numerical example

Exercise 4: Consider the credit balance problem, with

- $B_0 = \$1,000$, and
- $r = 0.12$.

Do the following.

- 1 Examine the evolution of the balance paying \$50 every month.
- 2 Interpret your examination relating to κ .

social feedback systems

Supply and demand can be described via feedback.

Proposal for a Topic of a Project:

Read the paper by Otto Mayr on “Adam Smith and the Concept of the Feedback System: Economic Thought and Technology in 18-th Century Britain” in *Technology and Culture*, Vol. 12, No. 1, 1971.

Adam Smith suggested two processes:

- 1 Workers compare various employments and stream into that employment which offers the greatest rewards.
- 2 The rewards diminish as the number of competing workers rises.

Do the following:

- Use feedback to describe the labor market.
- Apply your description to the academic labor market.
- Find data comparing mathematics and computer science.

software for control systems

Scilab and Xcos are open source software packages to design and analyze control systems.

Proposal for a Topic of a Project:

Visit `scilab.org`, download and install.

Do the following:

- Give an overview of the capabilities of the software.
- Reproduce one of the use cases listed on the web site.
- Write a report on your computational experiments.

summary and bibliography

We computed the amplitude gain and phase shift of a linear, causal, and time invariant filter, which lead to Bode plots.

Feedback was introduced with the credit card balance example.

We covered section 6 of Chapter 10 of our text book.

- Charles R. MacCluer:
Industrial Mathematics. Modeling in Industry, Science, and Government. Prentice Hall, 2000.

An introductory text book in control has many examples to engineering:

- Richard C. Dorf and Robert H. Bishop:
Modern Control Systems. Ninth Edition, Prentice Hall, 2001.